Zhengyang Yu

Software Engineering Student

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA
Expected Graduation: June 2028; GPA: 4.0/4.0
Relevant Coursework: Systems Programming, Computer Organization, Advanced Data Structures, Design and Analysis of Algorithms

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium [App Store] [GitHub]**, *Personal Project*
- Published a native SwiftUI companion for iPhone and Apple Watch on the App Store, providing multi-account quota monitoring, reset-credit reminders, background refresh, and a WidgetKit complication.
- Designed secure cross-device data flows using device authorization, Keychain, WatchConnectivity, URLSession, cached offline snapshots, and idempotent reset-credit redemption; verified with XCTest and physical-device testing.
- Built a companion ESP32-S3 C++/FreeRTOS implementation with adaptive HTTPS polling, touch UI, framebuffer graphics, IMU-driven water physics, and bounded persistent logs.

2025 **HTML to ePub Converter**, *Personal Project*
- Engineered a converter to transform HTML content into ePub format, implementing scalable parsing and metadata embedding.
- Enhanced reliability and efficiency through structured error handling and modular component design.

2024 **Home Assistant Presence Plugin**, *Personal Project*
- Built a Home Assistant plugin detecting real-time room occupancy using entry/exit sensor data.
- Designed event-driven architecture integrating Python scripts and RESTful APIs for smart lighting automation.

Skills

Programming Swift, C++, Python, Java, C
Apple Platforms SwiftUI, watchOS, WidgetKit, WatchConnectivity, Keychain, XCTest
Systems FreeRTOS, REST APIs, HTTPS, concurrent and event-driven systems, structured logging
Engineering Algorithms, data structures, object-oriented design, software testing, debugging
AI Development Codex-assisted codebase exploration, implementation, refactoring, debugging, and verification
Tools Git, Docker, Linux, PlatformIO, Xcode, VS Code

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch
Supported community members with computer troubleshooting and taught basic software skills.

2024 Tutor, CTN (Community Technology Network)
Provided one-on-one tutoring to help adults improve digital literacy skills, including computer operation and internet usage.